PDE Ledger

Archive Ledger Skeleton

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Purpose and Scope

TODO. B3 will fill the rebuilt ledger purpose, source hierarchy, and scope boundaries.

0.1 The Role of This Ledger

Placeholder for the rebuilt ledger's role and audience.

0.2 Primary Source Hierarchy

Placeholder for the source hierarchy used by the rebuilt ledger.

0.3 Document Contract

Placeholder for the provenance, status, verification, and downstream-use contract.

0.4 Scope Boundaries

Placeholder for the limits of the rebuilt ledger.

0.5 Completeness Criterion

Placeholder for the completion standard used by later build phases.

How to Read This Ledger

TODO. B3 will fill the reader guide after the rebuilt stage and Part structure is set.

0.6 Layer Structure

Placeholder for the relationship between theorem-block prose, stage cards, notes, and audits.

0.7 Reading Paths

Placeholder for operational, verification, provenance, and authoring reading paths.

0.8 Citation Discipline

Placeholder for result anchors, theorem labels, equation labels, and stage provenance.

0.9 Stage Entries

Placeholder for the expected fields in future stage cards.

0.10 Gap Handling

Placeholder for demotion, splitting, open-gate, and failed-route handling.

Claim-Status Firewall

TODO. B3 will fill the rebuilt ledger's status rules and examples.

0.11 Allowed Status Tags

Status tag	Meaning
Exact	Placeholder.
Exact Within Closure	Placeholder.
Reduced / Controlled Reduction	Placeholder.
Effective Closure	Placeholder.
Numerically Located	Placeholder.
Open	Placeholder.

0.12 Audit-Status Vocabulary

Placeholder for audit evidence categories.

0.13 Status Inheritance

Placeholder for weakest-essential-input and status propagation rules.

0.14 Citation Language

Placeholder for safe status-aware citation wording.

Notation Firewall

TODO. B3 will fill notation rules after the rebuilt sectors and stage packets are fixed.

0.15 Global Anti-Ambiguity Rules

Placeholder for global notation safeguards.

0.16 Coordinate, Index, and Operator Conventions

Placeholder for coordinate, index, and operator conventions.

0.17 Field and Sector Variables

Placeholder for sector-specific variable conventions.

0.18 High-Risk Symbol Collision Table

Placeholder for symbol collision accounting.

0.19 Notation Checklist

Placeholder for future stage write-up checks.

Result Anchor Map

TODO. B3 will assign rebuilt result anchors after the Part and stage plan is settled.

0.20 Anchor Use Rules

Placeholder for stable citation-anchor rules.

0.21 Anchor Naming Convention

Placeholder for rebuilt anchor naming.

0.22 Top-Level Result Anchors

Anchor	Stage range	Citation target
No anchors assigned yet.		

0.23 Maintenance Checklist

Placeholder for anchor maintenance rules.

Dependency Map

TODO. B3 will fill the rebuilt dependency graph after stages are authored.

0.24 Arrow Types

Placeholder for dependency-arrow types.

0.25 Global Dependency Spine

Placeholder for the rebuilt ledger's dependency spine.

0.26 Top-Level Dependency Table

Block	Inputs	Outputs
No dependency blocks assigned yet.		

0.27 Revision Rules

Placeholder for dependency-map revision rules.

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Stage cards for Part III are collected in Appendix C.

Appendix A

Part I Stage Cards

A.1 Stage 004: GNLS Action and Dimensional Foundation

Part / anchor. Part I – The medium

Claim status. Exact Within Closure

Purpose. Fix the field content and $\{L, T, M\}$ dimensional dictionary of the GNLS parent action, derive the pin null-relation and healing scale, and land the base-system verdict `RETAIN_L_T_M`.

Status. Exact Within Closure. The dimensional dictionary, sound-speed and healing anchors, pin null-relation, and base-system verdict are exact symbolic results; the honest landing carries named residuals (mass non-emergence and the $\{L, T\}$-conversion gaps), deferred to pathA_21/22 and not repaired here.

Parent action and field content. The 4D-bulk medium is a GNLS condensate with order parameter ψ, number density $\rho = |\psi|^2$, and EOS $P = K\rho^5$ (from $U(\rho) = \frac{K}{4}\rho^5$):

$$\mathcal{L} \sim i\hbar\,\psi^*\partial_t\psi - \frac{\hbar^2}{2m_{\mathrm{GNLS}}}\,|\nabla\psi|^2 - U(\rho).$$

Dimensional dictionary (two tiers). **Primitive inputs.** $[\hbar] = ML^2T^{-1}$, $[m_{\mathrm{GNLS}}] = M$, $[\rho_0] = L^{-4}$ (state datum) — posted and consistency-checked against action usage. **Derived by composition.** $[\psi] = L^{-2}$, $[\rho] = L^{-4}$, $[K] = ML^{18}T^{-2}$, forced by action homogeneity across all 17 harness checks (14 foundation, including the bulk continuity equation, plus 3 $\{L, T\}$-representation gates).

Derived anchors. **Sound speed and healing length.**

$$c_{s0}^2 = \frac{5K\rho_0^4}{m_{\mathrm{GNLS}}}, \quad [c_{s0}] = LT^{-1}; \qquad h_0 = \frac{5K}{4}\rho_0^4 = \frac{m_{\mathrm{GNLS}}c_{s0}^2}{4}, \quad \xi_h = \frac{\sqrt{2}\,\hbar}{m_{\mathrm{GNLS}}c_{s0}}.$$

Pin null-relation. Four pins $\{a, c_{s0}, \hbar, m_{\mathrm{GNLS}}\}$ on three base dimensions give rank 3, nullity 1:

$$a = \frac{\hbar}{m_{\mathrm{GNLS}}c_{s0}}, \qquad a/\xi_h = 1/\sqrt{2}.$$

Verdict. **RETAIN_L_T_M.** Computed from two conjuncts: the $\{L, T\}$-rejection gate (dropping M leaves ≥ 1 non-dimensionless target, needing M-bearing conversion factors LT^2M, L^2T^2M, L^2T^2) *and* the mass fork (m_{GNLS} explicit of dimension M; m_{defect} not derived, `INFLOW_MASS_SOURCE_MISSING`). The $\{L, T\}$ projection is a natural-unit representation only, not a base-system change.

Carried residuals (first-class, not repaired). `INFLOW_MASS_SOURCE_MISSING` (m_{defect} emergence deferred to pathA_21; $\hbar J/c_\gamma^2 = M$ is dimensional-only), `LT_R_norm_gate_fails_without_new_conversion_factor` (`REJECTS_TRUE_LT_BASE`) and the two 4D/3D R-norm conversion gaps (pathA_22), plus `A_PIN_IS_BRANCH_MOMENT_NO` `EOS_FROM_GNLS_FACTOR`, `NO_NET_ACCRETION_BC_UNDERIVED`, `M_TO_G_UNIFICATION`, `SCALE_MAP_INPUTS`.

A.2 Stage 005: EOS Sound Speed and Light/Sound Ratio as a Free Knob

Part / anchor. Part I – The medium

Claim status. Reduced / Controlled Reduction

Purpose. Derive the phonon sound speed from the imposed EOS, pin the three velocity scales and the $c = c_\gamma$ wave-sector ceiling, and land the light/sound ratio $\lambda_\gamma = c_\gamma / c_s$ as a free calibration input (`C_GAMMA_RATIO_UNDERDETERMINED`).

Status. Reduced / Controlled Reduction. The sound-speed law and the $c = c_\gamma$ ceiling are exact symbolic results *within the imposed EOS closure*; the honest headline is the CALIBRATED landing: $\lambda_\gamma = c_\gamma / c_s$ is *unpinned by the parent action* – a free calibration input, with the tail $(c/c_s)^3 = \lambda_\gamma^3$ carried symbolic (not set to 1). Consuming stage: it cites `ledger_stage004` (I-1)'s dictionary and `EOS_FROM_GNLS_FACTOR` handoff (h_0, ξ_h) and owns the sound-speed derivation I-1 deferred here.

Sound speed from the EOS. From the imposed $P = K\rho^5$ (`EOS_CLOSURE_IMPOSED`) and the barotropic definition, by symbolic differentiation:

$$c_s^2(\rho) = \frac{1}{m_{\mathrm{GNLS}}}\frac{dP}{d\rho} = \frac{5K\rho^4}{m_{\mathrm{GNLS}}}, \quad [c_s] = LT^{-1}, \qquad \frac{d\ln c_s}{d\ln\rho} = 2.$$

The factor 5 falls out of the derivative. EARNED relative to the imposed EOS.

Three speeds and the $c = c_\gamma$ ceiling. **Velocity scales.** $v_b = (\hbar/m_{\mathrm{GNLS}})\nabla\theta$, c_s, c_γ, each of dimension LT^{-1} (v_b's pin consumed from I-1). **Ceiling (derived from the wave sector).** For the trapped mode $\omega = c_\gamma\sqrt{k^2 + k_\perp^2}$,

$$\frac{d\omega}{dk} = \frac{c_\gamma k}{\sqrt{k^2 + k_\perp^2}}, \qquad c_\gamma^2 - \left(\frac{d\omega}{dk}\right)^2 = \frac{c_\gamma^2 k_\perp^2}{k^2 + k_\perp^2} \geq 0,$$

strictly positive for $k_\perp \neq 0$: the group speed is bounded by c_γ and approaches it. The bound-mode clock $\omega_0 = c_\gamma k_\perp$ gives the boosted internal rate ω_0/γ (time dilation) with no $E = mc^2$ premise.

Coupled principal symbol (pathA_20b support). On the neutralized homogeneous background the coupled symbol block-diagonalizes: the Madelung phonon determinant $\det M = \hbar(\omega^2 - c_{s0}^2 k^2)$ with $c_{s0}^2 = 5K\rho_0^4/m_{\rm GNLS}$ falling out via $h' = 5K\rho_0^3$, the gauge block $C_E\omega^2 - C_Bk^2 = C_E(\omega^2 - c_{\rm bulk}^2 k^2)$ ($c_{\rm bulk}^2 = C_B/C_E$), and $\det P = \det M \cdot P_T^2$. The off-diagonal current/London terms are lower order and do not set the cone (`BULK_PRINCIPAL_TRANSVERSE_BRANCH_ESTABLISHED`); $[c_s] = [c_\gamma]$ is labeled *non-evidentiary* for equality.

The free-knob landing and its negative control. C_GAMMA_RATIO_UNDERDETERMINED. Computed, not asserted. With $C_E, C_B, K, \rho_0, m_{\rm GNLS}$ independent, the residual $C_B/C_E - 5K\rho_0^4/m_{\rm GNLS}$ is not identically zero and no source equation is present, so `forced_equals_valid` = `False` $\Rightarrow$ `FORCED_EQUALITY_REJECTED_WITHOUT_SOURCE_EQUATION`. It is *reversible*: inserting the identity $C_B \to 5K\rho_0^4 C_E/m_{\rm GNLS}$ flips it to `C_GAMMA_EQUALS_C_S` – so `EQUALS` is reachable only via an actually-inserted source equation the action does not supply. Two-layer sharpening: bulk `C_GAMMA_BULK_UNDERDETERMINED`, brane `C_GAMMA_RATIO_STILL_UNDERDETERMINED`; if $c_{\rm bulk}$ is ρ_0-independent then $\lambda_\gamma \propto \rho_0^{-2}$ ($d\ln\lambda_\gamma/d\ln\rho_0 = -2$).

Carried residuals (first-class, not repaired). `EOS_CLOSURE_IMPOSED`; `C_GAMMA_RATIO_UNDERDETERMINED`; `STATIONARY_PROFILE_UNDERDETERMINED_BY_BRANCH_DATA`; `NO_NET_ACCRETION_BC_UNDERIVED`; `HBAR_PROVENANCE_U` `HBAR_FREE_SUBSTRATE_RELATION_MISSING`; `H_2PI_RATE_CLASSIFICATION_UNDERDETERMINED`; `BULK_METRIC_SPEED_` `PARENT_METRIC_ACOUSTIC_IDENTIFICATION_MISSING`; `BRANE_ZERO_MODE_REDUCTION_UNDERIVED`; `BRANE_PHOTON_CONE_REQUIRES_PROFILE`. The mass bridge $\hbar J/c_\gamma^2 = M$ is a labeled dimensional candidate only, not a $m_{\rm defect}$ derivation.

A.3 Stage 006: Two-Phase Material-State Ontology (Order Field χ_B)

Part / anchor. Part I – The medium

Claim status. Exact Within Closure

Purpose. Formalize the brane/bulk ontology as a POSTULATED two-phase action – one conserved medium, order field $\chi_B \in [0,1]$; brane = ordered shear-supporting phase, bulk = the same medium destructured, throat = phase-conversion site – dimensionally classified (`ACTION_SPECIFIED_CLASSIFIED`), with the recovery reduction to the frozen old-ledger $S_{\rm leak}$ verified assert-zero and the θ-as-Maxwell-φ no-go carried able-to-fail.

Status. Exact Within Closure. **The wall is made explicit as a postulated field – it is NOT derived from the one medium.** Every term of the free-energy functional and order-state balance is POSTULATED and labeled (13 pinned modeling choices P1–P13); the EARNED content is exact *within that imposed closure*: structural-closure identities, single-kink wall admission, and the recovery reduction against the frozen firewall. Cross-engine agreement proves consistent encoding of the same postulated F, not derivation. Consuming stage: cites `ledger_stage004` (dictionary, single-well U), `ledger_stage005` (c_{s0}^2), and `ledger_stage003` ($c_\gamma^2 = \mu_R/\rho_{br}$, $C_J = -J\rho_{B0}$, $B_{\rm eff} = \rho_{B0}^2/\chi_c$, the second-class$\Rightarrow$stray-DOF classification rule) with exact-value citation-integrity checks.

The postulated action and balance.

$$F = \int d^4X \Big[\tfrac{1}{2} m_{\rm GNLS} n |\mathbf{u}|^2 + U(n) + f_B(\chi_B) + \tfrac{\kappa_B}{2}|\nabla_4\chi_B|^2 + \chi_B f_{\rm shear} + f_{\rm throat} + f_{\rm mix}\Big],$$

$$\partial_t n + \nabla_4 \cdot (n\mathbf{u}) = 0, \qquad \partial_t(\chi_B n) + \nabla_4 \cdot (\chi_B n\mathbf{u} + \mathbf{J}_\chi) = n\Gamma_B, \quad [\Gamma_B] = T^{-1}.$$

Key pins: kinetic term carries the constituent mass (`KINETIC_MASS_FACTOR_PINNED`; $[n] = L^{-4}$); double-well $f_B = a_B\chi_B^2(1-\chi_B)^2$ with minima $\{0,1\}$, n-independent (`SAME_DENSITY_DEGENERACY_POSTULATED`) – required because the cited $U(\rho) = (K/4)\rho^5$ is *single*-well; MacCullagh-form shear gate on the displacement curl; $\chi_B \perp P$ (`POSTULATED_ANISOTROPY` dims-only); $\mathbf{J}_\chi = 0$ default; global returns $R_0 = -M_0$, $R_1 = -D_1$ printed as postulates, not asserted locally; throat = phase-conversion ontology, not a solve. Every integrand term audited to $ML^{-2}T^{-2}$; adjunct rows $(\mu_\chi, M_\chi, P_{\text{order}}, M_n)$ dimension-pinned, with the energy ledger fixed to the variationally consistent $P_{\text{order}} = \int \mu_\chi D_t\chi_B \, d^4X$ (`HANDOFF_P_ORDER_N_PLACEMENT_CORRECTED`).

Structural closure and wall admission (EARNED within closure). The advective form $D_t\chi_B = \Gamma_B - (1/n)\nabla_4 \cdot \mathbf{J}_\chi$, the disorder complement for $n_D = (1-\chi_B)n$, and the Γ_B-cancelling sum reproducing total conservation are derived by real derivative expansion in both engines. The static EL equation $\kappa_B\chi_B'' = f_B'$ admits the kink $\chi_B = \frac{1}{2}(1 + \tanh(w/2\delta))$ with solved width $\delta = \sqrt{\kappa_B/2a_B}$ and surface tension $\sigma_{\text{wall}} = \sqrt{2a_B\kappa_B}/6$, $[\sigma_{\text{wall}}] = ML^{-1}T^{-2}$. **Slab caveat (carried).** Bulk on both sides makes the brane a finite kink–antikink *slab*; the double-well selects no slab width – W_{slab} is an un-earned input mapping onto the old ledger's L/a self-selection item (sim-deferred). Wall admission $\neq$ slab stability.

Recovery reduction (EARNED within closure; the frozen firewall). Projection with unit-normalized $W(w)$ and in-engine IBP gives the two-source law $\partial_t\rho_B + \nabla_3 \cdot \mathbf{J}_B = S_{\text{flux}} + S_{\text{convert}}$ with the general $\chi_B(w), \Gamma_B(w), J_\chi^w(w)$ computationally live; the in-engine limit $\chi_B \to 1, \Gamma_B \to 0, \mathbf{J}_\chi \to 0$ reduces it assert-zero to the *independently constructed* frozen target $S_{\text{leak}} = -[Wj^w]_\partial + \int W'j^w \, dw$, $j^w = nu^w$ (stage_243), including the stage_244 Gaussian one-mode closed form $S_{\text{leak}} = \sqrt{2}\,\mu_w\rho_0\mathcal{E}_0/(2\sqrt{\pi}\lambda^3)$ re-derived by direct integration (`RECOVERY_REDUCTION_VERIFIED`). Conditionality computed: $\chi_B = c \neq 1$, $\Gamma_B \neq 0$, $J_\chi^w \neq 0$ each leave nonzero computed residuals – the limit does real work.

The θ-as-Maxwell-φ no-go (carried FATAL_FLAW). A stable order-parameter phase has positive gradient energy ($\kappa_{\text{phase}} > 0$) $\Rightarrow$ $K_\theta = -\kappa_{\text{phase}} < 0$ in the Lagrangian; Maxwell's electric square needs $K_\theta = C_J^2/\rho_{br} > 0$ – opposite signs (exact predicate). The constraint bracket $\{\Phi_1, \Phi_2\} = k^2(J^2\rho_{B0}^2 + \kappa_{\text{phase}}\rho_{br})/\rho_{br}$ has its zero locus solved in-engine at exactly $K_\theta = +J^2\rho_{B0}^2/\rho_{br}$: via the consumed stage_003 rule the provenance branch lands `FAIL_CAUCHY_STRAY_LONGITUDINAL` (finite $B_{\text{eff}} = \rho_{B0}^2/\chi_c$), the $B_{\text{eff}} = 0$ branch `FAIL_C5_LONGITUDINAL_ZERO_MODE`, and the tuned fixture $(K_\theta = J^2\rho_{B0}^2/\rho_{br}, B_{\text{eff}} = 0, m_\theta^2 = 0)$ `C5_RESOLVED_MAXWELL_BY_TUNING` – flagged `BY_TUNING`, never `WITH_PROVENANCE` (flag computed from provenance booleans; detunings re-fire the pathA_36 control tokens). The only provenance sign-flip ($\kappa_{\text{phase}} < 0$) is a Lifshitz instability, $k_*^2 = -\kappa_{\text{phase}}/2\kappa_4 > 0 \iff \kappa_{\text{phase}} < 0$ – the already-killed pathA_25 wall (`FAIL_LIGHT_STARVED` lineage). The transverse sector is untouched: the ε-parametrized dispersion derived in-engine gives $\omega^2 = (\mu_R/\varepsilon)k^2$; $\varepsilon = \rho_{br}$ reproduces the consumed c_γ^2 (`PASS_TRANSVERSE_UNDISTURBED`), $\varepsilon \neq \rho_{br}$ fires `FAIL_TRANSVERSE_DISTURBED`. **This stage does not earn light** – the no-go is the point.

Drift and carried items (first-class, not repaired). `DRIFT(6)` computed: $\{\chi_B$; a_B; κ_B; α_{aniso}; Γ_B; gating struct (rung-W reconciliation printed; `THETA_BRANCH_DEAD_NOT_ADMITTED`; ρ_{B0}, χ_c cross-referenced to the pathA_41 Part-VI drift trio, not double-counted). Carried: `SECOND_MEDIUM_DRIFT` lineage, the slab width W_{slab}, Gate-L $\delta w = u_w$ translation Goldstone (deferred), $\mathbf{J}_\chi \neq 0$, $f_{\text{throat}}/f_{\text{mix}}$ placeholders, dynamics/energy-ledger adjunct. Two source-prose dimensional defects pinned and ablation-documented (kinetic mass factor; P_{order} n-placement).

A.4 Stage 007: Shear-Surface G0 Freeze (Frozen Medium Action, Drift Ledger, DOF)

Part / anchor. Part I – The medium (closing Part-I stage)

Claim status. Exact Within Closure

Purpose. Formal home of the pathA_35 G0 shear-surface *constitutive* freeze: the frozen brane/light action ($L_{\mathrm{Mac}}+L_{Pu}+L_{u_w}$ under the Gaussian profile $g_\ell(w)$), hash-guarded (`T0_SHEAR_FROZEN(d9520d3819c3)`), with the honest drift ledger `SECOND_MEDIUM_DRIFT_AT_FREEZE(11)` computed from an enumerated table and the flat-brane linear DOF$=8$ rank-computed.

Status. Exact Within Closure. **The 11 freeze inputs are POSTULATED/CALIBRATED – the freeze freezes *terms*, not gate answers.** The EARNED content is exact within that declared closure: (i) *freeze fidelity* – both engines re-extract the canonical `freeze-action` blocks by byte-exact fence-parsing (never fixed offsets) and assert SHA-256 equality (G0 `d9520d3819c3f7...`, T0 `8fa41ac51e88...` byte-embedded); (ii) the *flat-brane linear DOF* $= 8$ computed from rank/nullity bookkeeping (in-plane u^a curl-transverse 2 + kinetic–curl 1; T0 P^i tangent 3 + radial 1; u_w 1; C5 φ **0** – the absent longitudinal partner the later C5 crux attacks); (iii) the *dimensional firewall* over every frozen term. This stage computes **no** gate verdict (Gate-L excluded; exposure names carried as prose provenance only) and does **not** earn light (Stage 003 derives the two transverse photons at $c_\gamma^2 = \mu_R/\rho_{br}$ from this stage's frozen L_{Mac}).

The frozen action. Kept (0 new): the GNLS parent action and the T0 polar-OP action (byte-for-byte, hash-embedded). Frozen brane blocks under $S_{\mathrm{brane}} = \int dt\, d^4X\, g_\ell(w)\,[L_{\mathrm{Mac}} + L_{Pu} + L_{u_w}]$:

$$L_{\mathrm{Mac}} = \tfrac{1}{2}\rho_{br}(\partial_t u^a)^2 - \tfrac{1}{2}\mu_R\,\Omega_u^a\Omega_u^a, \qquad L_{Pu} = -\lambda_{Pu}\,\varpi_a\,\Omega_u^a, \quad \varpi_a := (\hat{w}\times P_\parallel)_a,$$

$$L_{u_w} = \tfrac{1}{2}\rho_{br}(\partial_t u_w)^2 - \tfrac{1}{2}\rho_{br}\Omega_w^2 u_w^2, \qquad g_\ell(w) = \frac{e^{-(w/\ell_g)^2}}{\sqrt{\pi}\,\ell_g}, \quad \int g_\ell\, dw = 1 \text{ (derived)}.$$

Preserved verbatim: *L_{Pu} re-admits the ϵ-contracted/chiral class excluded by T0* and requires the conceded axis $\hat{w}$ – a structural-postulate cost, not a free choice.

The drift ledger (computed). Both engines enumerate the 11 and *compute* the subcounts and the verdict string (the source gate's hardcoded literals are retired): **4 constants** $\rho_{br}\,[ML^{-3}]$, $\mu_R\,[ML^{-1}T^{-2}]$, $\lambda_{Pu}\,[ML^{-1}T^{-2}]$, $\Omega_w\,[T^{-1}]$ (table dims asserted against the firewall targets); **1 function** $g_\ell(w;\ell_g)$ (fixed Gaussian, one width, target-blind admission); **6 structural postulates** (conceded $\hat{w}$+wall; free-slip u^a; P^i reused as Cosserat reservoir; massless spin-wave baseline; the parity-even P–u operator; no C5 φ). New-field content (u^a 3 + u_w 1 = 4 DOF) is counted *separately.* **Erratum (2026-07-04), carried first-class: the 11 STANDS** – no ρ_{br} overcount (pathA_25's $\varrho_{br}[\rho]$ belongs to the closed density-smectic candidate, `OUT_OF_ACTIVE_NG5`); $\{\rho_{br}, \mu_R\}$ carry the registered-pending Route-A reduction (`ROUTE_A_UNDERDETERMINED_MISSING_NONLINEAR_THROAT`, register R10); the honest cross-sector drift $\{\rho_{B0}, \chi_c, C_{hu}\}$ is a Part-VI (pathA_41) item, guarded in-engine against absorption into the 11.

Supersession and firewalls. Both facts asserted: the Stage 006 χ_B wall superseded the fixed-shape $g_\ell(w)$ as the *material-state* closure, while G0 *remains* the light-sector *constitutive* freeze (Stage 003 consumes L_{Mac} as-is) – a scope split (register R21), not a reduction. Notational firewall (register R22): μ_R (3D brane, $ML^{-1}T^{-2}$) and Stage 006's $\mu_R^{(4)}$ (4D density, $ML^{-2}T^{-2}$) are asserted dimensionally distinct, related only by the *pending* R17 projection $\mu_R = \int \chi_B \, \mu_R^{(4)} dw$ (dim-consistency asserted).

Verification. Dual-engine, independent routes (the source pair's `-compare` JSON payload mirror is retired): SymPy 96 PASS / Mathematica 94 PASS, both exit 0, CWD-independent. Able-to-fail teeth all fire: the five source-gate ablations (2 dimensional, 3 DOF each recomputing $7 \neq 8$), two hash teeth (in-block byte corruption; missing fence), five enumeration teeth (drop entry; miscategorize; inject ρ_{B0}; corrupt n pre-verdict; corrupt table dim), and the μ_R firewall tooth. Full tri-review on fresh agents: arbiter re-run, `FIDELITY_CLEAN` (block-by-block coverage diff, no dropped check), `ADVERSARIAL_CLEAN` (26-run mutation matrix: hash genuinely fence-parsed – one byte inside the block fails, one byte outside passes; dual-corruption of both engines fails both; no cross-engine comparison anywhere).

Appendix B

Part II Stage Cards

B.1 Stage 001: Solid-Angle and Second-Moment Primitives

Part / anchor. Part II – Gravity

Claim status. EXACT

Purpose. Derive the unit-sphere solid angles and normalized second moments used by the pathA_21c matter-stress force assembly.

Status. EXACT. This stage is closed geometry: no profile, normalization, or field-equation residual remains inside the stated claims.

Derivation. For $S^2 \subset \mathbb{R}^3$, take

$$n(\theta, \phi) = (\cos\theta, \sin\theta\cos\phi, \sin\theta\sin\phi), \qquad dS = \sin\theta\, d\phi\, d\theta.$$

Then **Omega2 solid angle.**

$$\Omega_2 = \int_0^\pi \int_0^{2\pi} \sin\theta\, d\phi\, d\theta = 4\pi.$$

With $n_1 = \cos\theta$ and $n_2 = \sin\theta\cos\phi$, **S2 second moments.**

$$\langle n_1^2 \rangle_{S^2} = \frac{1}{4\pi} \int_0^\pi \int_0^{2\pi} \cos^2\theta\, \sin\theta\, d\phi\, d\theta = \frac{1}{3}, \qquad \langle n_1 n_2 \rangle_{S^2} = 0.$$

The cross moment vanishes because $\int_0^{2\pi} \cos\phi\, d\phi = 0$.

For $S^3 \subset \mathbb{R}^4$, take

$$n(\chi, \theta, \phi) = (\cos\chi, \sin\chi\cos\theta, \sin\chi\sin\theta\cos\phi, \sin\chi\sin\theta\sin\phi), \qquad dS = \sin^2\chi\, \sin\theta\, d\phi\, d\theta\, d\chi.$$

Then **Omega3 solid angle.**

$$\Omega_3 = \int_0^\pi \int_0^\pi \int_0^{2\pi} \sin^2\chi\, \sin\theta\, d\phi\, d\theta\, d\chi = 2\pi^2.$$

With $n_1 = \cos\chi$ and $n_2 = \sin\chi\cos\theta$, **S3 second moments.**

$$\langle n_1^2 \rangle_{S^3} = \frac{1}{2\pi^2} \int_0^\pi \int_0^\pi \int_0^{2\pi} \cos^2\chi\, \sin^2\chi\, \sin\theta\, d\phi\, d\theta\, d\chi = \frac{1}{4}, \qquad \langle n_1 n_2 \rangle_{S^3} = 0.$$

The cross moment vanishes because $\int_0^\pi \cos\theta \sin\theta\, d\theta = 0$.

Physical use. $\Omega_2 = 4\pi$ and $\Omega_3 = 2\pi^2$ set the reduced r^{-2} and bulk R^{-3} Gauss normalizations. The second moments give the δ_{ij}/d angular average used in the convective factor $-(1+1/d)$ and the pressure factor $1/d$.

B.2 Stage 002: Matter-Stress Inter-Defect Force Assembly

Part / anchor. Part II – Gravity

Claim status. Reduced / Controlled Reduction

Purpose. Assemble the inter-defect force between two drain defects from the matter (Noether) stress, earning the bilinear Q_1Q_2 structure, the reduced-3D r^{-2} and bulk-4D R^{-3} power laws, and the attractive like-drain sign, consuming the Stage 001 geometry primitives.

Status. Reduced / Controlled Reduction. Three verdict layers are carried from the earned source: the leading matter-stress sign `FORCE_ATTRACTIVE_DERIVED` (target-blind); the full sign `SIGN_RESIDUAL_QUANTUM_VCONF_MAXWELL_PROFILE` (honest residual); acceptance `PASS_WITH_NAMED_RESIDUALS`. The bilinear structure, the power laws, and the attractive sign are earned (form + sign); the overall magnitude is a calibration knob; the full sign carries named, profile/sim-deferred residuals.

Derivation. The medium obeys $P = K\rho^5$, $h = (5K/4)\rho^4$, with the barotropic identity $dP/d\rho = \rho\,(dh/d\rho)$. The matter (Noether) stress in Madelung form is

$$\Pi_{ij} = m\rho\, v_i v_j + \delta_{ij} P(\rho) + \sigma^Q_{ij}, \qquad \sigma^Q_{ij} = \frac{\hbar^2}{4m}\Big[\frac{(\partial_i\rho)(\partial_j\rho)}{\rho} - \partial_i\partial_j\rho\Big],$$

with the quantum-stress divergence contributing no net far-field force at this order. A defect draining flux Q sets up, by Gauss's law, the inflow

$$v = -\frac{Q\,\hat{n}}{\Omega_{d-1}\, r^{\,d-1}},$$

where $\Omega_2 = 4\pi$ $(d = 3)$ and $\Omega_3 = 2\pi^2$ $(d = 4)$ are the Stage 001 solid angles. The Bernoulli cross-pressure of the two overlapping inflows is $\delta P = -m\rho\,(v_1\cdot v_2)$. Integrating the traction $\Pi_{ij}n_j$ over a control sphere around defect 2 and using the Stage 001 second moment $\langle n_i n_j\rangle = \delta_{ij}/d$ gives the convective factor $-(1+1/d)$ and the pressure factor $+1/d$, whose sum is the total flux $-\,mNQ_2v_1$ (the $1/d$ pieces cancel). The stationary force $F_{12} = -\oint \Pi_{ij}n_j\, dS$, with defect 1's own inflow substituted, yields the two lanes.

Reduced-3D force law.

$$F_{12} = -\frac{m\,N_3\,Q_1Q_2}{4\pi\, r_{12}^2}\,\hat{r}_{12}.$$

Bulk-4D force law.

$$F_{12} = -\frac{m\,\rho_4\,Q_1Q_2}{2\pi^2\, R_{12}^3}\,\hat{R}_{12}.$$

Attractive sign. Since $F \propto -Q_1Q_2$, like drains $(Q_1Q_2 > 0)$ attract and opposite signs repel; the sign is derived from the Gauss-drain / Bernoulli / surface-integral chain, not assumed.

Physical use and named residuals. Target-blind, the inter-defect force is bilinear in the drain strengths, falls as r^{-2} (reduced 3D) and R^{-3} (bulk 4D), and is attractive between like drains. Honest scope: the overall normalization (I_F/Θ_Q/branch-profile) is a *calibration* knob, not derived; and the full sign carries the `QUANTUM_STRESS_FAR_FIELD`, `VCONF_BODY_FORCE`, and `MAXWELL_Z_GAUGE_JEXT` residuals, which are profile/sim-deferred and first-class.

B.3 Stage 008: Monopole/Dipole Return Constraint Specification

Part / anchor. Part II – Gravity (first return/radiation stage)

Claim status. OPEN

Purpose. Render the old ledger's open-item #9 as an exact constraint specification: the raw $\ell = 0/1/2$ outgoing c_s-wave amplitudes and radiation orders, and the precise source moments any brane↔bulk return must cancel (`MONOPOLE_DIPOLE_RETURN_CONDITIONAL`). The formal home of the targets $R_0 = -M_0$, $R_{1,i} = -D_{1,i}$ consumed by Stages 009/010 (pathA_29) and 022/023 (pathA_34); register edge R23.

Status. OPEN – deliberately. **This is a verified CONSTRAINT-SPECIFICATION, not a falsifiable test of no-monopole-radiation** (the source gate's own scope caveat, carried verbatim-class). GR forbids monopole/dipole gravitational radiation; the drain breaks brane mass conservation, so suppression is conditional on a return that cancels the moments below. The cancellation condition is algebraic bookkeeping (an $x - x$ identity, labeled as such in both engines); `cancellation_possible` is a literal parameter flag printed as *SCOPE: parameter, not computed – track-3 decides.* The falsification lives downstream (Stages 009/010; the Gate-6 $Z_{0,\mathrm{ret}}/Z_{1,\mathrm{ret}}$ selector).

The DtN ladder (earned; scanned, not typed). From spherical Hankel closed forms $h_\ell = j_\ell + iy_\ell$, both engines form $\Lambda_\ell = k\, h'_\ell/h_\ell|_{z=ka}$, normalize the admittance by its static value, and *scan* for the first purely-imaginary k-coefficient:

$$p_{\mathrm{raw}} = 1, 3, 5; \qquad \text{kernels } i\frac{\omega}{c_S}a, \quad i\frac{\omega^3}{c_S^3}\frac{a^3}{2}, \quad i\frac{\omega^5}{c_S^5}\frac{a^5}{27},$$

$$A_{\mathrm{raw}}^{(\ell 0)} = \frac{iM_0 a\omega}{c_S}, \qquad A_{\mathrm{raw}}^{(\ell 1)} = \frac{iD_1 a^3\omega^3}{2c_S^3}, \qquad A_{\mathrm{raw}}^{(\ell 2)} = \frac{iQ_2 a^5\omega^5}{27c_S^5},$$

all symbolically nonzero; steady control $\lim_{\omega\to 0} A^{(\ell 0)} = 0$; dominance $p(\ell{=}0) < p(\ell{=}2)$ from the scanned powers.

Cancellation targets (bookkeeping, labeled). $R_0(\omega) = -M_0(\omega)$ with $M_0 = \int_{\mathrm{brane}} S_{\mathrm{leak}}\, d^3x$ (cancels $O(\omega^1)$); $R_{1,i}(\omega) = -D_{1,i}(\omega)$ with $D_{1,i} = \int x_i S_{\mathrm{leak}}\, d^3x + \int j_i\, d^3x$ including the carried odd wake (cancels $O(\omega^3)$). Conditions, not closure. The derivative outlet vertex $g_{W0} = \eta\omega$ is recorded as a `branch_assumption`, not verdict-bearing. Live-source consistency is anchored by direct integration of the frozen Stage-243/244 closed forms ($\int W' j^w dw = -\sqrt{2}\,\ell_w j_0/4$; $S_{\mathrm{leak}} = \sqrt{2}\,\mu_w\rho_0 E_0/(2\sqrt{\pi}\lambda^3)$) with the strict-recovery limits computed but *not* taken as a basis; Stage 006 owns the recovery-reduction derivation (boundary printed).

Verdict machinery and guards. The verdict is computed from computed booleans plus the literal flag; the synthetic control (True, False, False) $\rightarrow$ `MONOPOLE_RADIATION_UNAVOIDABLE` proves the rung reachable. The nine source controls are carried honestly: four computed, five printed as declarations (not counted checks), with the source's framing – guards against obvious tautologies, not evidence of suppression-vs-unavoidable. Consumed from Stage 005: $c_s^2 = 5K\rho^4/m_{\text{GNLS}}$ (cited, exact-value integrity check $5 \cdot \frac{1}{500} \cdot (\sqrt{10})^4 = 1$); frozen slice $G = c = c_s = 1$, $K_{\text{eos}} = 1/500$, (a^*, L^*) exact rationals, cited. Q_2 is exported as a **free anchor** (not derived $\ell = 2$ physics) for Stage 026.

Verification. Dual-engine, independent routes, zero file I/O (the source pair's JSON/YAML engine bridge is retired): SymPy 54 PASS / Mathematica 57 PASS (3 = arity self-checks), both exit 0, CWD-independent. Eight able-to-fail teeth fire (Hankel-form corruption trips the scan; kernel-coefficient, moment-kill, bookkeeping-integrity, verdict-machinery, anchor-prefactor, steady-limit, consumed-law corruptions). Full tri-review on fresh agents: `FIDELITY_CLEAN` (coverage-diff, math re-derived by hand) and `ADVERSARIAL_CLEAN` (14/14 mutation matrix; dual-corruption fails both engines; arity scan incl. planted-mismatch detection), followed by a nit remediation (tally hygiene, restored recovery-slice observations, pipeline-routed tooth) re-verified clean.

B.4 Stage 009: Flat-Slab Return Residual (Check A)

Part / anchor. Part II – Gravity (brane$\leftrightarrow$bulk return, Check A)

Claim status. Exact Within Closure

Purpose. Derive, on the postulated flat-slab family, what a finite DC-sink brane$\leftrightarrow$bulk return actually delivers against the Stage-008 cancellation targets: the return transfer $T_\ell(\omega) = \alpha_\ell e^{i\omega\, 2d/c_s}$, the bounded residual orders $p_{\text{res}}(\ell{=}0) = 1$, $p_{\text{res}}(\ell{=}1) = 3$, and the signed local source accounting $Z = -M_0\varepsilon_0/(1+\varepsilon_0)$ – the Check-A component of `RETURN_RESIDUAL_PREDICTION` (Check-B localization = Stage 010).

Status. Exact Within Closure – exact *within the postulated slab family* (brane at $w = 0$, return/absorber at $w = d$; the geometry is postulated, the return response is derived on it). The v3 labeling is carried exactly: $Z < 0$ (drain admissibility) is the **premise** (`Z_is_premise=true`); the formula $Z = -M_0\varepsilon_0/(1+\varepsilon_0)$ is **accounting** (channel sum $Z_{\text{throat}} + Z_{\text{return}}$, sign certificate computed). Open-item #9 is *sharpened, not closed*; the headline is printed only as the Check-A component with the Stage-010 qualifier.

The derived return response. From the bidirectional Helmholtz basis $\Phi_\ell = A_\ell e^{i\omega w/c_s} + B_\ell e^{-i\omega w/c_s}$, the round-trip transit time is *solved* ($\tau = 2d/c_S$, asserted via $e^{i\omega\tau} \equiv$ phase). DC continuity fractions $\alpha_\ell = 1/(1{+}\varepsilon_\ell)$ with independent per-channel transmissions $\varepsilon_0, \varepsilon_1 > 0$; steady circulation balances ($J_{\text{leak}} = J_{\text{return}} + J_{\text{neighbor}}$, both moments) assert-zero. A genuine ω-order scan computes $\nu_\ell = \operatorname{ord}_\omega(1 - T_\ell) = 0$ – a finite DC sink leaves the ω^0 deviation $1 - T_\ell(0) = \varepsilon_\ell/(1+\varepsilon_\ell)$ – so the residual amplitudes $A_{\text{res}} = \text{kernel}_\ell \cdot \text{moment} \cdot (1 - T_\ell)$ sit at **computed** orders $p_{\text{res}} = p_{\text{raw}} + \nu_\ell = 1, 3$: a *bounded monopole/dipole c_s-radiation residual tied to the drain strength* – the falsifiable departure from GR. The Check-A classification booleans are computed from the orders.

Reversibility (per-channel, computed). At $\varepsilon_0 \to 0^+$: $T_0 \to e^{i\omega\tau}$, strict $\nu_0 = 1 \Rightarrow p_{\text{res},0} = 2$, and $Z \to 0$ (a perfect-return channel has no drain). At $\varepsilon_1 \to 0^+$: strict $\nu_1 = 1$ computed *directly from the $\ell = 1$ limit* (the source reused ν_0; the reshape computes it independently) $\Rightarrow p_{\text{res},1} = 4$. The prediction is contingent on the transmissions, not baked in. Consumed from Stage 008 (cited, dual-site integrity-checked): the three radiation kernels; Q_2/kernel$_2$ inert (`T2_applied=false`). Register: rows $\{d, \varepsilon_0, \varepsilon_1\}$ (slab-family parameters; ε's FREE-UNREDUCED on the Gate-6 route) + edge R24.

Verification. Dual-engine, independent routes, zero file I/O – the source pair's JSON-digest bridge, its runtime YAML reads of pathA_28 outputs, and its SHA-256 trace bookkeeping are all retired. SymPy 48 PASS / Mathematica 52 PASS, exit 0, CWD-independent. Eight able-to-fail teeth fire. Full tri-review on fresh agents: `FIDELITY_CLEAN` (all values hand-re-derived; 009/010 boundary clean); the adversarial leg (16-mutant matrix) **caught a genuine rig-class defect** – the consumed-kernel citation-integrity block was set-then-compare-to-self (a single-source corruption passed) – remediated to dual-site integrity (acceptance: all four one-site corruptions now fail in both engines) plus tally-hygiene nits, then re-verified clean. The `Off[Limit::alimv]` suppression was unmasked and shown benign (every silenced limit's result independently asserted).

B.5 Stage 010: Slab Localization $p = 2$ with Reachable NOGO (Check B)

Part / anchor. Part II – Gravity (brane↔bulk return, Check B; completes the pathA_29 fold)

Claim status. EXACT WITHIN CLOSURE

Purpose. Show, on the postulated flat-slab family, that Newtonian $1/r^2$ gravity SURVIVES the finite slab: both admissible DC-sink completions carry a normalizable m=0 transverse zero mode whose 3D-radial Green function gives flow $\propto 1/r^2$ ($p = 2$); the anti-localizing warp control gives $p = 3$ and the same classifier returns `RETURN_NOGO` (the able-to-fail companion). This computes Check B and emits the COMPLETED joint verdict `RETURN_RESIDUAL_PREDICTION` = (Check A cited, Stage 009) ∧ (Check B computed here).

Status. EXACT WITHIN CLOSURE – exact *within the postulated slab family* (brane at $w = 0$, return/absorber at $w = d$; the geometry is postulated, localization is derived on it). The gravity-range ($1/r^2$) leg passes inside the localizing flat-slab family because both DC-sink completions give $p = 2$; the joint verdict `RETURN_RESIDUAL_PREDICTION` is completed here, with `RETURN_NOGO` genuinely reachable. Open-item #9 is *sharpened, not closed*: localization holding for the FAMILY is not the family being SELECTED by dynamics (R19/W_{slab}); the return-cancellation targets (R23) remain deferred.

The localizing zero mode and $p = 2$. On $0 \le w \le d$ each admissible completion is a transverse eigenproblem $f'' + m^2 f = 0$ with a normalizable constant m=0 zero mode ($f_0 = 1/\sqrt{d}$, $\int_0^d f_0^2\, dw = 1$): the *destructuring/absorbing* (compact cell, Neumann) and *Bloch* (q=0 periodic) cells, with gapped first modes at $m_1 = \pi/d$, $2\pi/d$. The 3D-radial zero-mode equation $g'' + (2/r)g' + ((\omega/c_s)^2 - m^2)g = 0$ is solved by genuine `dsolve` (dynamic route: solved first, outgoing spherical branch selected via C_1/C_2 – a boundary SELECTION, normalization = compact overlap $1/d$ – then $\omega \to 0$); the limit Green $1/(4\pi dr)$ gives flow $1/(4\pi dr^2)$ and *extracted* exponent $p = 2$. The static route ($\omega = 0$ first) gives the same $1/(4\pi dr)$ and $p = 2$; the two routes agree on both the exponent and the Green ($dynamic_{\omega\to 0} = static$). The massive/gapped solve gives a Yukawa $e^{-m_1 r}/r$ – only the m=0 zero mode sets the far field. Both completions agree $p_{\text{abs}} = p_{\text{bloch}} = 2$.

Able-to-fail: the counterfactual and the NOGO warp. Multiplying the solved Green by r^{-4} ($\to 1/r^5$) yields radial-operator residual $5/(\pi d r^7) \neq 0$ – REJECTED, proving the solve has teeth. The mandatory anti-localizing warp $\mu(w) = e^{2k_{\mathrm{warp}} w}$ has a *non-normalizable* zero mode ($\int_0^\infty \to \infty$, computed) and a continuum Green $1/(4\pi r^2)$ giving $p_{\mathrm{delocalizing}} = 3$; the same classifier returns `RETURN_NOGO`. The classifier verdict is a function of the computed exponents (and the cited `quadrupole_survives`), never a stamp; `RETURN_NOGO` is reachable if the return delocalizes OR the quadrupole channel does not survive.

Consumed (cited, dual-site integrity + exact anchor). From Stage 008: $p_{\mathrm{raw}}(\ell{=}2) = 5 \Rightarrow$ `quadrupole_survives` (finite quadrupole order; T_2 not applied, `T2_applied=false`, no $\ell{=}2$ recompute). From Stage 009: the Check-A component `A_residual_pass=true` (joint-composition print only). Register: *zero new counted knobs*; k_{warp} is a control-construction symbol (tracked, not counted); edge R25 (localization $p = 2$ EARNED-within-family, able-to-fail via the warp NOGO – does not discharge R19 or R23).

Verification. Dual-engine, independent routes, zero file I/O – the source pair's JSON-digest bridge and all SHA-256 trace/`structure_id`/`expr_digest` bookkeeping are retired; the `.wl` solves the radial ODEs, the normalizations, the warp divergence, and the classifier natively (own `DSolve` basis, own `Limit`/`Integrate`), with an arity self-check and no silenced messages. SymPy 71 PASS / Mathematica 81 PASS (81 = 71 shared + 10 arity self-checks), exit 0, CWD-independent. Eight able-to-fail teeth fire. Full tri-review on fresh agents: `FIDELITY_CLEAN` (all values hand-re-derived; 009/010 boundary clean – Check-A cited not recomputed) and `ADVERSARIAL_CLEAN` (22-mutant matrix; a hardcoded-wrong-Green mutant confirmed the counterfactual guard is genuine, and the classifier genuinely reaches `RETURN_NOGO`). The adversarial leg's one substantive find – a *coordinated both-sites* corruption of the Stage-008 citation to another finite integer escaped – was folded to Stage-009 parity by an exact-value anchor, and re-verified (`REVERIFY_CLEAN`) as the sole gate closing that escape.

B.6 Stage 011: Frozen-Reduction Certificate (L_s Helmholtz) (Check II-G1a)

Part / anchor. Part II – Gravity (brane↔bulk return; the frozen-throat longitudinal reduction; opens the pathA_30 D/N fold, completed by Stage 012)

Claim status. Reduced / Controlled Reduction

Purpose. Certify that, on the postulated straight finite throat with the wall FROZEN ($R = R_0,\ \eta = 0$) and the matter perturbation left dynamic ($\propto e^{-i\omega t}$), the reduced longitudinal operator collapses to the constant-coefficient Helmholtz resonator operator $L_s\psi = \psi''(s) + (\omega/c_S)^2\psi(s) = 0$, with $c_S^2 = 5K\rho_*^4/m$ on $s \in [0, L_0]$ (cap $R_0(L_0) = 0$) — *produced* by a per-term reduction (projection measure plus every intruding coefficient computed with its vanishing/deferral condition), NOT typed — together with the reduction certificate: the validity window under which the collapse is exact.

Status. Reduced / Controlled Reduction — a controlled reduction with an explicit validity certificate. This stage carries the reduction-certificate component of the joint pathA_30 verdict `DN_UNITTEST_BC_DEPENDENT`; the D/N determinant, the DtN $-(\omega/c_S)\tan(L_0\omega/c_S)$, the half-shifted pole ladder, the Robin counterfactual, and the `BC_DEPENDENT` landing (*bc_derivation_emitted* = false, a banked calibration input) are Stage 012's; the cut is drawn at the general-solution `dsolve` (never called here). The wave speed is *banked* (Part I edge R1, Stage 005, at the straight-reference density ρ_*), so this stage adds NO reduction of its own: its contributions are the two *structural* edges R26 (the validity record) and R27 (the $\xi \neq \ell_c$ firewall), which discharge no knob.

The produced operator (the de-rig). The reduced operator is *assembled* from dimensionally-consistent terms,

$$L_s \;=\; \psi'' \;+\; M\,\psi' \;+\; N\,\psi \;-\; B\,\frac{\hbar^2}{4m^2c_S^2}\,\psi'''',$$

with $M = \frac{d}{ds}\log\sqrt{\gamma_0}$ ($[M] = L^{-1}$), the ψ-coefficient $N = (\omega/c_s(s))^2$ with $c_s(s)^2 = 5K\rho_0(s)^4/m$ ($[N] = L^{-2}$), and $B \in \{0,1\}$ the Bogoliubov inclusion flag — the k^4 correction being a genuine *fourth-derivative* intrusion (from the identity $\psi'' + (\omega/c_S)^2\psi - (\hbar^2/4m^2c_S^2)\psi'''' = 0$ for the full BdG dispersion), NOT a coefficient shift. The three source checks are replaced by genuine computations that can each fail: `operator_is_helmholtz` is expr_equal($L_s -$ ideal$, 0$) on the *assembled* L_s (not two hand-typed twins); `speed_is_cs` is *extracted* from the ψ-coefficient of L_s ($= (\omega/c_S)^2$, bulk, not wall/healing-renormalized, not literal `True`); `domain_is_L0` is *solved* from the pinch-off $R_0(s) = 0$ on a POSTULATED monotone taper $R_0(s) = R_{\mathrm{mouth}}(1 - s/L_0)$ (root $= L_0$, R_{mouth}-independent; a labeled SELECTION, not `L0==L0`); and `unsuppressed_operator_intrusion` is *computed* from the assembly (not hardwired `False`) and gates the verdict.

The reduction certificate (validity window). Under the frozen background ($\rho_0(s) = \rho_*$, $\sqrt{\gamma_0} = A_{\perp 0}$ constant, $A_{M0} = 0$) every intruding term is computed to vanish or is deferred: the projection-measure coefficient $M = 0$; $\rho_0' = 0 \Rightarrow c_s' = 0 \Rightarrow N = (\omega/c_S)^2$ (the bulk sound speed); $\delta V_{\mathrm{conf}} = 0$; the background quantum potential $Q(\rho_0) = -\frac{\hbar^2}{2m}\partial_s^2\sqrt{\rho_0}/\sqrt{\rho_0}$ gives $\partial_s Q = 0$; and the BdG term is deferred ($B = 0$) only under $k\xi \ll 1$ ($\xi = \hbar/(mc_s)$), with smallness witness $\hbar^2k^2/(4m^2c_S^2) = (k\xi/2)^2$. Thus L_s is const-coefficient Helmholtz *conditional on* $\{\rho_0'/\rho_0 = 0,\ \sqrt{\gamma_0}\ \text{const},\ \delta V_{\mathrm{conf}} = 0,\ \nabla Q = 0,\ k\xi \ll 1\}$ (edge R26); the BdG k^4 term is DEFERRED, not dropped unconditionally. The healing length $\xi = \hbar/(mc_s)$ and the confinement length ℓ_c (in $V_{\mathrm{wall}}(\Sigma/\ell_c)$) are kept as DISTINCT symbols (edge R27, the firewall analogous to R22's $\mu_R \neq \mu_R^{(4)}$); ℓ_c is INERT here ($\delta V_{\mathrm{conf}} = 0$).

Able-to-fail teeth. Eight teeth, mutations on copies: a nonconstant $\sqrt{\gamma_0}$ makes $M \neq 0$; a retained BdG ($B = 1$) injects the ψ'''' term; a nonuniform $\rho_0(s)$ makes N s-dependent (each $\to$ `FAIL_OPERATOR_INTRUSION`); a wall/healing renormalization $\to$ `FAIL_WRONG_SPEED`; a corrupted taper root $\to$ `FAIL_WRONG_DOMAIN`; an $\ell_c \to \xi$ conflation trips the firewall; a one-site R1 exponent corruption trips the dual-site integrity (a coordinated both-site drift is caught by the explicit frozen-export anchor); and a corrupt-$[K]$ mutation flips $[c_S^2]$ to $(3, 0, -2) \to$ `DN_UNITTEST_FAIL_DIMENSIONAL`. A dedicated witness-path tooth (nonzero δV_{conf} with $M = 0,\ N = \text{bulk},\ B = 0$) drives the verdict to `FAIL_OPERATOR_INTRUSION` through the intrusion flag alone, making the flag independently load-bearing.

Consumed + dimensional leg. The speed law $c_s^2 = 5K\rho^4/m_{\mathrm{GNLS}}$ is CITED (Part I edge R1, Stage 005) evaluated at ρ_*, integrity-checked dual-site (literal `5*K*rho**4/m` vs. the EOS route $\partial_\rho(K\rho^5)/m$) plus an explicit frozen-export anchor; the EOS exponent-5 closure $P = K\rho^5$ stays `IMPOSED`. The c_S^2 dimensional leg uses the $[K] = [P] - 5[\rho]$ chain in four spatial dimensions: $[K] = (18, 1, -2)$, $[c_S^2] = (2, 0, -2)$ (order L, M, T); the corrupt-$[K]$ probe flips it to $(3, 0, -2)$. L_0 (throat depth) is POSTULATED `ACTION`-geometry (like Stage 009's d, not a medium constant); zero new counted knobs.

Verification. Dual-engine, independent routes, zero file I/O — the source pair's scratch-YAML/`_sympy_exprs.wl` export, the Mathematica-YAML re-read, and the `expression_digest/engine_agreement` plumbing are retired; the `.wl` derives the certificate, the L_s assembly, the speed extraction, the domain solve, and the dimensional leg natively (own `D/Solve/Coefficient/Simplify`), with an arity self-check and no silenced messages; the transfer matrix, if present, is only an operator-nature cross-check (no DtN/pole/Robin — Stage 012). SymPy 61 PASS / Mathematica 71 PASS (71 = 61 shared + 8 arity self-checks + 2 native trig-basis confirmations), exit 0, CWD-independent. Full tri-review on fresh agents: `FIDELITY_CLEAN` (all values hand-re-derived; the three booleans confirmed genuinely PRODUCED; the 011/012 boundary clean) and `ADVERSARIAL_CLEAN` (the de-rig proven able-to-fail — operator via $M/N/B$, speed via renormalization, domain via taper — and the BdG confirmed a real fourth-order term). Three teeth/label nits were remediated (the firewall tooth made genuine, a witness-path tooth added so the intrusion flag is independently load-bearing, the arity labels corrected) and re-verified `REVERIFY_CLEAN`.

B.7 Stage 012: DtN Pole Ladder & Robin Falsifier ($Z_{00} = -(\omega/c_S)\tan$) (Check II-G1b)

Part / anchor. Part II – Gravity (brane↔bulk return; the frozen-throat resonator response; COMPLETES the pathA_30 D/N fold opened by Stage 011)

Claim status. Exact Within Closure

Purpose. On Stage 011's cited frozen constant-coefficient Helmholtz operator $L_s\psi = \psi'' + (\omega/c_S)^2\psi = 0$ on $s \in [0, L_0]$, pose and solve the Dirichlet/Neumann boundary-value problem: derive the outward-mouth Dirichlet-to-Neumann map $Z_{00} = -(\omega/c_S)\tan(L_0\omega/c_S)$, its half-shifted resonance ladder $\omega_n = \pi c_S(n + \frac{1}{2})/L_0$, its round-trip $R_{\mathrm{rt}} = 1$, and the Robin counterfactual that proves the D/N determination is not hardcodable — landing the joint verdict at `DN_UNITTEST_BC_DEPENDENT` because the boundary condition itself is imposed ($bc\_derivation\_emitted$ = false, a banked calibration input).

Status. Exact Within Closure — the DtN, ladder, round trip, and Robin counterfactual are exact algebra *within the imposed-BC closure*. This stage COMPLETES the joint pathA_30 verdict `DN_UNITTEST_BC_DEPENDENT`: Stage 011 carried the reduction-certificate component (`REDUCTION_CERTIFIED`), and Stage 012 adds the D/N ladder plus the `BC_DEPENDENT` landing. It is a near-pure *bridge-strip* (contrast Stage 011's de-rig): the physics is already genuinely computed in the source and the `.wl` transfer-matrix route already carries it, so the reshape severs the scratch-YAML bridge and keeps the native content. The operator, domain, and speed are *consumed* from Stage 011 (dual-site citation-integrity); the stage opens at the general-solution `dsolve`. Its register contribution is the single *imposed* edge R28 (the D/N boundary is a banked calibration input), which discharges no knob.

The D/N boundary-value problem and the DtN. The `dsolve` of the cited L_s gives $\psi = C_1\sin(\omega s/c_S) + C_2\cos(\omega s/c_S)$. Imposing the Dirichlet mouth trace $\psi(0) = \psi_M$ and the Neumann cap $\psi'(L_0) = 0$, the 2×2 coefficient matrix (determinant $-\omega\cos(L_0\omega/c_S)/c_S$) is solved by `LUsolve`, and the outward-mouth admittance is

$$Z_{00} \;=\; -\frac{\partial_s\psi|_0}{\psi|_0} \;=\; -\frac{\omega}{c_S}\,\tan\Big(\frac{L_0\omega}{c_S}\Big).$$

`dtn_matches_target` compares this *derived* Z_{00} (from the `LUsolve`) against the typed target $-k\tan(kL_0)$ — a genuine derived-vs-typed comparison, not an $X \equiv X$: perturbing the Neumann RHS recomputes the derived DtN and the comparison fails.

Ladder, static limit, round trip. The DtN denominator's $\cos(L_0\omega/c_S)$ factor gives the pole equation $\cos(L_0\omega/c_S) = 0$, whose roots are the *half-shifted* ladder $\omega_n = \pi c_S(n+\frac{1}{2})/L_0$; `halfshift` is computed as *pole_residual* $= 0$. The small-ω series $-L_0\omega^2/c_S^2 - L_0^3\omega^4/(3c_S^4) + O(\omega^6)$ is the leading static compliance (distinct from the DC limit $\lim_{\omega\to 0^+} Z_{00} = 0$; no separate static solve). With Dirichlet/Neumann reflections $r_D = -1$, $r_N = +1$, the round-trip factor $-\exp(2iL_0\omega/c_S)$ evaluates to $R_{\mathrm{rt}} = 1$ ($\varphi_0 = 0 \bmod 2\pi$) on the ladder.

The Robin falsifier. Without a counterfactual the D/N determination is hardcodable. The Robin cap $\partial_s\psi - \alpha\psi = 0$ interpolates Neumann ($\alpha \to 0$) and Dirichlet ($\alpha \to \infty$); its DtN reads off *robin_denominator_core* $= k\cos(kL_0) - \alpha\sin(kL_0)$. The `counterfactual_guard` has exactly six members, each a *computed residual* (not a stamped truthiness): `robin_determinant_emitted` tied to the computed Robin determinant *robin_det* $\equiv -$*robin_denominator_core*; `recovers_DN_at_alpha0` ($\alpha \to 0$ recovers $-(\omega/c_S)\tan$); `recovers_DD_at_alpha_inf` ($\alpha \to \infty$ recovers the D/D DtN $k\cot(kL_0)$); `halfshift_destroyed_for_DD` (the D/D denominator $\sin(kL_0)$ gives the *integer* ladder $\pi c_S j/L_0$, not the half-shifted one); `numeric_alpha_distinct` ($\alpha = 2/L_0$ distinct from both limits, exact rationals); and `dtn_mismatch`. The artifact *dd_zero_mode_removable* $= 1/L_0$ is held out of the guard. Breaking *robin_denominator_core* flips the recomputed `recovers_DD_at_alpha_inf` and is caught ($\to$`FAIL_COUNTERFACTUAL`).

Imposed BC (honest scope) and the completed joint verdict. The D/N boundary pair is IMPOSED (*bc_provenance* $=$ imposed); the explicit mouth/cap V_{wall} gradient derivation is *not* emitted (*bc_derivation_emitted* $=$ false), so the verdict lands at `DN_UNITTEST_BC_DEPENDENT` rather than `DN_UNITTEST_PASS` (edge R28; earning `PASS` is a deferred upgrade, not fabricated here). The 012-scoped verdict is a function of the computed rungs (`FAIL_DIMENSIONAL` $\to$ `FAIL_POLE_LADDER` $\to$ `FAIL_COUNTERFACTUAL` $\to$ the `BC_DEPENDENT` landing), and the joint verdict is printed as `DN_UNITTEST_BC_DEPENDENT` $=$ (011: `REDUCTION_CERTIFIED`, cited) $\wedge$ (012: DtN ladder + `BC_DEPENDENT` landing, computed here), not typed as 012-earned.

Consumed + dimensional leg. The frozen operator L_s, domain $[0, L_0]$, and speed c_S are CONSUMED from Stage 011 with dual-site citation-integrity: for L_s, site A the typed export form and site B the *null-space reconstruction* from the `dsolve`'d pair $\{\sin, \cos\}$ (posit $y'' + ay' + by$, solve for $(a, b) = (0, (\omega/c_S)^2)$) — a genuinely independent construction (a $\sin/\cos \to \sinh/\cosh$ corruption recovers $b = -(\omega/c_S)^2$ and fires); for $c_S^2 = 5K\rho^4/m$ (R1 at ρ_*), the literal vs the EOS route $\partial_\rho(K\rho^5)/m$. The 012 dimensional legs use the $[K] = [P] - 5[\rho]$ chain: [tan_argument] $= [kL_0] = 1$ (dimensionless), $[Z_{00}$_prefactor$] = [-k] = L^{-1}$; the corrupt-$[K]$ probe propagates $[c_S^2] \to [c_S] \to [k]$, flipping both legs $\to$ `DN_UNITTEST_FAIL_DIMENSIONAL`. α (Robin cap admittance, $[\alpha] = L^{-1}$) is a control-construction symbol; zero new counted knobs.

Verification. Dual-engine, independent routes, zero file I/O — the source pair's scratch-YAML/`_sympy_exprs.wl` export, the Mathematica-YAML re-read, and the `expression_digest/engine_agreement` plumbing are retired; the `.wl` KEEPS its transfer-matrix route as the independent engine (propagate the mouth state through `transferMatrix`$[L_0]$, impose the Neumann cap, read Z_{00} — a different decomposition from the `.py` `LUsolve`, computed agreement), preserving its native `robinAlpha0==dtnTransfer`/`robinAlphaInf==ddTransfer` self-checks, with an arity self-check and no load-bearing message silenced. SymPy 84 PASS / Mathematica 90 PASS (the +6 = the `.wl` arity block net of two SymPy-only DtN-detail checks), exit 0, CWD-independent. Full tri-review on fresh agents: `FIDELITY_CLEAN` (all values hand-re-derived; the six-member Robin guard each a genuine residual; `dtn_matches_target` genuine derived-vs-typed; the L_s null-space dual-site independent; the consume-from-011 boundary clean) and `ADVERSARIAL_CLEAN` (16-mutant matrix — the guard un-stampable, the sin / cos → sinh / cosh corruption fires in both engines, even a hardcoded DtN target is caught, all four verdict rungs reachable, arity clean). Eight able-to-fail teeth, mutations on copies.

Appendix C

Part III Stage Cards

C.1 Stage 003: Transverse Photon Pair and the Stray Longitudinal Mode

Part / anchor. Part III – Light

Claim status. Open

Purpose. Test whether the shear-surface brane carries light of the Maxwell kind. The transverse question earns a clean result – two massless photons at $c_\gamma^2 = \mu_R/\rho_{br}$; the longitudinal question earns a characterized departure – one stray second-class mode where Maxwell has a gauge-removed zero, with the tuned Maxwell locus reachable only by tuning.

Status. Open. FAIL-headline stage: the top-line verdict is a `FAIL_*`, but the earned content is the headline and the FAIL is the first-class characterized landing (no softening). *Earned headline:* the brane carries *light* – two massless transverse photons, with (for the first time in the program) $c_\gamma^2 = \mu_R/\rho_{br}$; verdict `PASS_TRANSVERSE_UNDISTURBED`. *Top-line departure:* `FAIL_CAUCHY_STRAY_LONGITUDINAL` – on the provenance-fixed single-medium branch the $(\nabla\cdot u) + \theta$ sector carries one stray propagating longitudinal DOF, a Dirac–Bergmann *second-class* pair rather than a Maxwell first-class Gauss gauge-removal. *Reachability:* the tuned Maxwell locus ($K_\theta = C_J^2/\rho_{br}$, $B_{\text{eff}} = 0$, $m_\theta^2 = 0$) is reachable and emits `C5_RESOLVED_MAXWELL_BY_TUNING` – so the FAIL is not hardcoded – but `BY_TUNING`, not `WITH_PROVENANCE`.

Derivation. The primitive brane Lagrangian (no pre-completed square used as input) is

$$L = \tfrac{1}{2}\rho_{br}(\partial_t u)^2 - \tfrac{1}{2}\mu_R(\nabla\times u)^2 - \tfrac{1}{2}B(\nabla\cdot u)^2 + J(\partial_t\theta)\,\delta\rho_B + \tfrac{1}{2}K_\theta(\nabla\theta)^2.$$

Transverse (earned). For a polarization $\perp$ $\mathbf{k}$, $\nabla\cdot u = 0$ and the θ couplings vanish, so $L_T = \tfrac{1}{2}\rho_{br}(\partial_t u_T)^2 - \tfrac{1}{2}\mu_R k^2 u_T^2$, giving

$$\omega^2 = \frac{\mu_R}{\rho_{br}}\,k^2, \qquad c_\gamma^2 = \frac{\mu_R}{\rho_{br}},$$

two polarizations $\Rightarrow$ physical_dof = 2, massless. The speed is able-to-fail: $\epsilon = 2\rho_{br}$ shifts it to $\mu_R/(2\rho_{br})$ (`FAIL_TRANSVERSE_DISTURBED`). **Longitudinal (departure).** Number conservation slaves $\delta\rho_B = -\rho_{B0}\nabla\cdot u$; integration by parts gives the Maxwell-form cross-term with the *derived* sign $C_J = -J\rho_{B0}$, and the finite conjugate-density stiffness the Cauchy modulus $B_{\text{eff}} = \rho_{B0}^2/\chi_c$. Since $\partial_t\theta$ enters only through the first-order cross-term, the momenta are $p_u = \rho_{br}\partial_t u_L$, $\pi_\theta = Jk\rho_{B0}u_L$, forcing the primary constraint $\Phi_1 = \pi_\theta - Jk\rho_{B0}u_L$; preservation gives $\Phi_2 = -k(Jp_u\rho_{B0} + k\kappa_{\rm phase}\rho_{br}\theta)/\rho_{br}$, and

$$\{\Phi_1, \Phi_2\} = \frac{k^2(J^2\rho_{B0}^2 + \kappa_{\rm phase}\rho_{br})}{\rho_{br}} \neq 0$$

$\Rightarrow$ second-class pair (0 first-class, 2 second-class), $N_{\rm phys} = (4-2)/2 = 1$. The finite-stiffness pole $\omega^2 = k^2\kappa_{\rm phase}\rho_{B0}^2/[\chi_c(J^2\rho_{B0}^2 + \kappa_{\rm phase}\rho_{br})]$ has positive residue and a bounded reduced Hamiltonian: a real stray longitudinal mode, not a ghost and not a gauge-removal.

Physical use, fidelity, and the departure. The brane carries light: two transverse photons at $c_\gamma^2 = \mu_R/\rho_{br}$ from the medium's own moduli (light = in-plane brane shear). Honest scope: the order-parameter phase θ does *not*, on its frozen definitions, supply a gauge-removed Maxwell scalar – it leaves one stray second-class longitudinal mode; the obstruction is pinned to the Cauchy modulus $B_{\rm eff} = \rho_{B0}^2/\chi_c$ plus the conventional phase-gradient sign $K_\theta = -\kappa_{\rm phase} < 0$ (vs the electric sign $K_\theta = C_J^2/\rho_{br} > 0$). Fidelity: the decisive sign $C_J = -J\rho_{B0}$ is *derived* here by an in-script integration by parts (the earned source had asserted it), closing that stage's known `FIDELITY_ISSUES`; the conventional K_θ sign is a labeled postulate whose verdict-dependence is exercised by the reachable Maxwell-locus control. Resolution of the stray mode is calibrated / sim-deferred; μ_R, ρ_{br} are calibrated inputs.

Appendix D

Reproducibility Map

D.1 Purpose of This Appendix

Placeholder for the rebuilt ledger's reproducibility rules.

D.2 Reproducibility Layers

Layer	What it controls	Minimum evidence required
No reproducibility layers populated yet.		

D.3 Canonical Source Registry

Placeholder for source artifacts.

D.4 Primary Symbolic-Audit Registry

Placeholder for symbolic audit artifacts.

D.5 Repository Freeze Protocol

Placeholder for future release-freeze requirements.

Appendix E

Source File Index

E.1 Purpose of This Appendix

Placeholder for the rebuilt ledger's source-file index.

E.2 Source Classes

Placeholder for source classes.

E.3 Generated Ledger Files

File family	Role
`paper/frontmatter/*.tex`	Placeholder governance chapters.
`paper/parts/*.tex`	Future theorem-block chapters.
`paper/stages/stage_NNN.tex`	Future per-stage cards.
`scripts/ledger_stageNNN_*.py`	Future SymPy audit scripts.
`mathematica/ledger_stageNNN_*.wl`	Future Mathematica audit scripts.

E.4 Precedence and Conflict-Resolution Rules

Placeholder for source precedence and conflict-resolution rules.

Appendix F

Layer-2 Fit-Insertion Index

F.1 Generated Coverage Summary

Item	Count
Phase-A candidates	0
Audited candidates	0

F.2 Phase-A Candidate Rows

No candidate rows exist in the empty ledger skeleton.

F.3 Visible Grouping and Exclusion Accounting

No grouping or exclusion accounting exists yet.

Appendix G

Layer-2 Parameter-Provenance Audit

G.1 Generated Coverage Summary

Item	Count
Top-level provenance parameter records	0
Canonical candidates	0

G.2 Published-Target Benchmark Sources

No benchmark rows exist in the empty ledger skeleton.

G.3 Parameter Genealogy Rows

No parameter genealogy rows exist yet.

G.4 Visible Alias Accounting

No alias accounting exists yet.

Appendix H

Fill Workflow and Editorial Rules

H.1 Purpose of This Appendix

Placeholder for the rebuilt ledger fill workflow.

H.2 What Filled Means

Placeholder for completion criteria.

H.3 Governing Maps Used During Filling

Placeholder for status, notation, anchor, dependency, source, and reproducibility controls.

H.4 Stage Appendix Fill Protocol

Placeholder for future stage-card authoring rules.

H.5 Theorem-Block Fill Protocol

Placeholder for future Part authoring rules.

H.6 Minimal Build and Static-Check Workflow

1. Confirm every retained `\input` target exists.
2. Run the SymPy and Mathematica audit runners.
3. Build `paper/pde_ledger.tex`.